

Assignment – 1.3

DAY 3

Day 3: RAM and Storage Theory (30 min): Types of RAM (DDR3, DDR4, DDR5) and storage (HDD vs. SSD, NVMe). Practical (30 min): Remove and reinstall RAM and storage drives.

Task - 1

Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.

Feature	DDR3	DDR4	DDR5
Generation	Older Gen	Newer than DDR3	Latest from all
Voltage	Approx. 1.5	1.2	1.1
Bandwidth	less	More than ddr3	Maximum bandwidth
Data transfer rates	800-2133 MT/S	1600-3200MT/S	4800-8800MT/S
Uses	Older PC	Laptops	Newer PC
Desktop Pin	240	288	288

Task 3

List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.

Feature	HDD	SATA SSD	NVMe
Technology	Magnetic Storage	Flash Memory	Flash Memory
Speed	Slower	Faster	Fastest
Moving Parts	Yes	No	No
Interface	Sata	SATA	PCIe
Protocol	SATA/AHCI	SATA/AHCI	NVME
Noise	Make some noise	Silent	Silent
Uses	Bulk Storage	OS, Windows	Higher Gaming PC
Costs	Lower per GB	Higher than HDD	More Costly

Task-2

Open your laptop or desktop (with power disconnected), locate the RAM slots, and take a clear photo of the installed RAM module. Reinstall the RAM module and verify that your system boots up successfully



Task-4

Remove the storage drive (HDD or SSD) from your system, take a photo of the connector type (SATA, M.2, or NVMe), and reinstall the drive. Confirm that your device recognizes the storage after reassembly

Video link: <https://youtu.be/w8W1xTLEJ6Y?si=T13KOF0LlxBaH0Ai>